

Mammalia Raccoon Proximity Network Analysis

<https://github.com/aiyshwary/Raccoon-Network-Analysis>

OVERVIEW

A graph-based epidemiological study modelling rabies transmission risk in urban raccoon populations through centrality measures, supporting public health and wildlife management decisions.

IMPACT

Constructed and analyzed a raccoon proximity network using real-world contact data, applying centrality measures to rank individuals and quantify potential rabies spread pathways within urban populations.

KEY DETAILS

1. Built a weighted proximity network from the mammalia-raccoon-proximity dataset using Python and NetworkX.
2. Computed degree, betweenness, closeness, and eigenvector centrality to rank individuals by transmission risk.
3. Identified superspreader nodes whose removal would most disrupt disease propagation chains.
4. Visualized network topology and centrality distributions to communicate findings clearly.
5. Results provide evidence-based recommendations for targeted wildlife intervention strategies.
6. Analyzed a dense 24-node, 226-edge proximity graph (~82% density) derived from 1,997 weighted temporal contacts.
7. Computed clustering coefficients and aggregated per-node contact strength (total edge weight) to quantify high-risk hubs; the highest total contact weight (956,341) was observed for a central node.
8. Rendered the network using Kamada-Kawai force-directed layout (Matplotlib) to reveal spatial clustering and high-risk hubs.